TIGLAD SAVRA

MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

ME20: DESIGN AND GRAPHICS — TIGTRAY

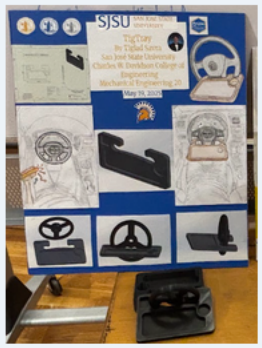


Figure 1. Final class presentation summarizing the problem, concept evolution, and prototype.



Figure 2. Hand sketch translating the drive-through eating problem into a steering-wheel tray concept.



Figure 5. 3D-printed prototype showing the assembled wheel and tray with verified part fit.

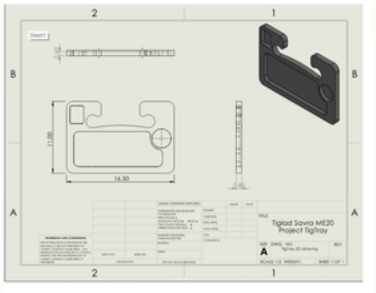


Figure 3. SolidWorks drawing with dimensions, section views, and drawing notes for the tray component.

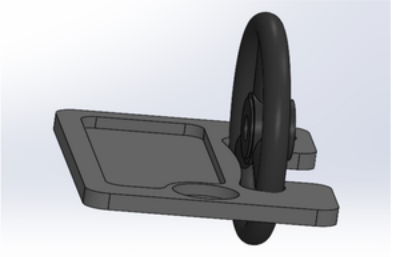


Figure 4. SolidWorks assembly showing the steering wheel and TigTray interface before fabrication.

DESIGN TARGET

Compact cars & sedans: 14–15.5 in (350–395 mm) steering-wheel range

Targeted for a stable, comfortable, and adaptable fit with future sizing expansion in mind.

What?

- Designed **TigTray,** a steering-wheel-mounted food tray that addresses a practical drive-through problem: keeping food stable and accessible in a parked car without spills, mess, or unnecessary heat loss.
- Translated a hand-drawn problem statement into a two-part product concept sized around the **14–15.5 in (350–395 mm)** steering-wheel range for compact cars and sedans.
- Carried the final project from concept and market rationale through **SolidWorks modeling,** assembly, dimensional analysis, drawings, and scaled fabrication.

How?

- Modeled individual parts in **SolidWorks,** assembled them with mating features, and produced a dimensioned drawing to communicate form, fit, and manufacturing intent.
- Used iterative **3D printing** in IdeaMaker to compare tray concepts, confirm scaled geometry, and verify that the wheel and tray **mated** as designed.
- Applied **design-for-use** thinking to create a stable eating surface, integrated cup-holder/storage features, compact packaging, and an interface adaptable to a target steering-wheel range.

Results

- Earned an **A** by delivering a presentation-ready prototype that connected **concept development, CAD, dimensioning, 3D printing,** and physical fit verification.
- Demonstrated design iteration by moving from sketch to CAD to printed parts, using the prototype to expose fit and usability considerations before future expansion.
- Built practical mechanical-design skills in **SolidWorks, assembly, tolerancing, prototyping,** and communicating a product from an idea into a manufacturable form.

TIGLAD SAVRA

MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

ENGR-10: INTRODUCTION TO ENGINEERING — WIND TURBINE

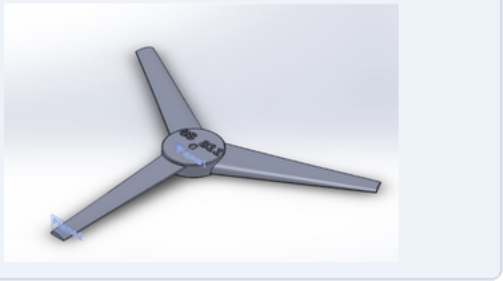


Figure 1. SolidWorks blade concept 1 showing the baseline three-blade rotor geometry.

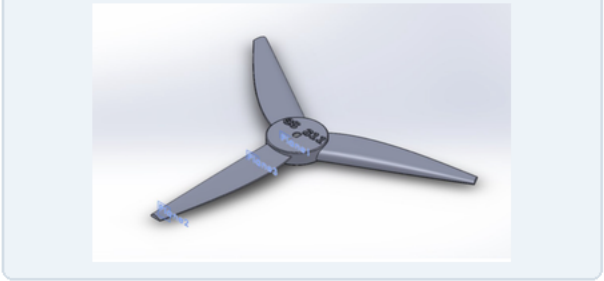


Figure 2. Revised SolidWorks blade concept with a tapered and curved profile for wind applications.

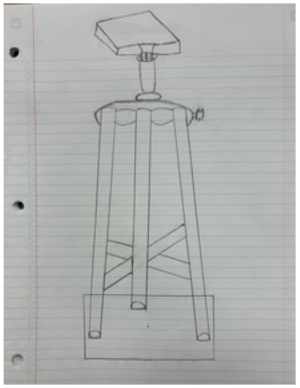


Figure 3. Initial structural sketch defining the tripod, adjustable center post, top platform, and rotor placement.



Figure 4. Pre-final tripod prototype showing the adjustable three-leg base and wood support platform.



Figure 5. Final turbine test setup with the mounted motor/generator, wiring, and electrical control components operating together.

DESIGN / TEST FOCUS

Three-blade rotor • adjustable height • 360° orientation • load distribution • live electrical output

What?

- Developed a sustainable, adjustable wind-turbine system for **ENGR-10: Introduction to Engineering,** combining a three-blade rotor, motor/generator, structural frame, and electrical test setup within the course dimensions.
- Researched three-blade rotors as a cost-effective, efficient baseline, then designed two blade geometries in **SolidWorks** by varying diameter, tip profile, and curvature for wind applications.
- Selected a repurposed tripod as the structural platform because its three-leg base distributed load, supported height adjustment, and enabled controlled rotation.

How?

- Applied **statics** and **mechanics of materials** to reason through bending, torsion, load paths, and stress distribution from the rotor and motor into the platform, center post, and tripod legs.
- Iterated blade and structure prototypes through **SolidWorks** and **3D printing;** used screws, padded feet, tape, and a thick wood plate to improve stability and protect contact points.
- Integrated adjustable height, 360° orientation, and a geared **locking mechanism** so the turbine could be positioned for testing and held at a controlled setting.

Results

- Earned an **A** by delivering a working turbine prototype that produced electrical output while the motor/generator, wiring, and controls operated together.
- Achieved one of the strongest stability results in the SJSU class comparison during stress/strain testing, validating the tripod-based load distribution and rigid top platform.
- Strengthened **SolidWorks, mechanical design, structural analysis, rapid prototyping, sustainability,** and collaborative test-and-improve skills applicable to energy systems and product development.

TIGLAD SAVRA

MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

ENGR-10: INTRODUCTION TO ENGINEERING — MOBILE ROBOT

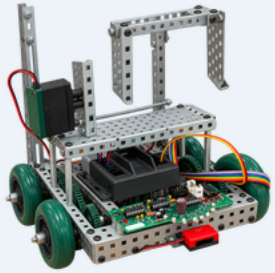


Figure 1. Early mechanical build showing the wheeled frame, vertical arm structure, motors, and IR-board integration.

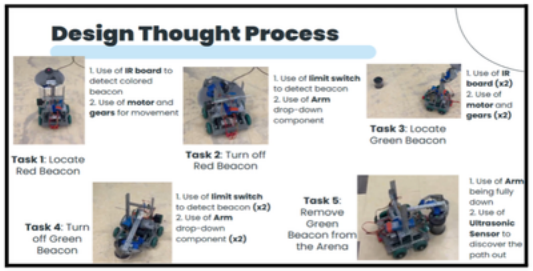


Figure 2. Design thought process mapping beacon detection, button actuation, object pickup, and the timed exit sequence.

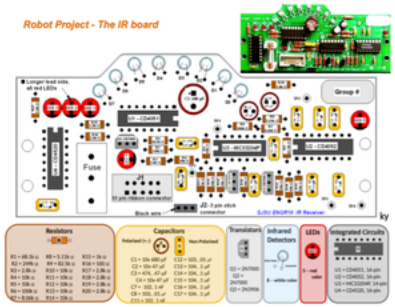


Figure 3. IR-board component layout identifying resistors, capacitors, transistors, LEDs, detectors, and integrated circuits before soldering.

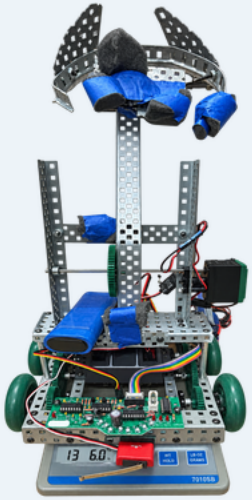


Figure 4. Final mobile robot on a scale, showing the integrated frame, arm, wheels, IR board, wiring, and battery.

SYSTEM CONSTRAINTS

Weight • height • mobility • timed task • soldered IR sensing • RobotC control

Designed as a portable system that could sense, move, actuate, retrieve, and deliver under time pressure.

What?

- Built a portable robot for the **ENGR-10 lab final** that completed a timed sequence: detect and turn off beacons, pick up an object with an arm/lever, and deliver it.
- Integrated a wheeled frame, motor-driven arm, relays, IR sensing, and onboard electronics within **weight, height,** and mobility constraints.
- Worked as a team to balance reach, speed, stability, and repeatability without a prescribed mechanical solution.

How?

- Tuned motor placement, gearing, lever geometry, and end-effector alignment so the robot could contact, lift, carry, and release the task object.
- Built and soldered the **IR board,** then connected sensors, relays, motors, and wiring into a reliable detection-and-actuation path.
- Programmed the timed sequence in **RobotC** and iterated code, circuitry, arm placement, and motion timing through trial testing.
- Applied **Dynamics** to moving loads and reaction forces so the arm stayed **secure,** remained **movable,** and prevented **tipping** during actuation.

Results

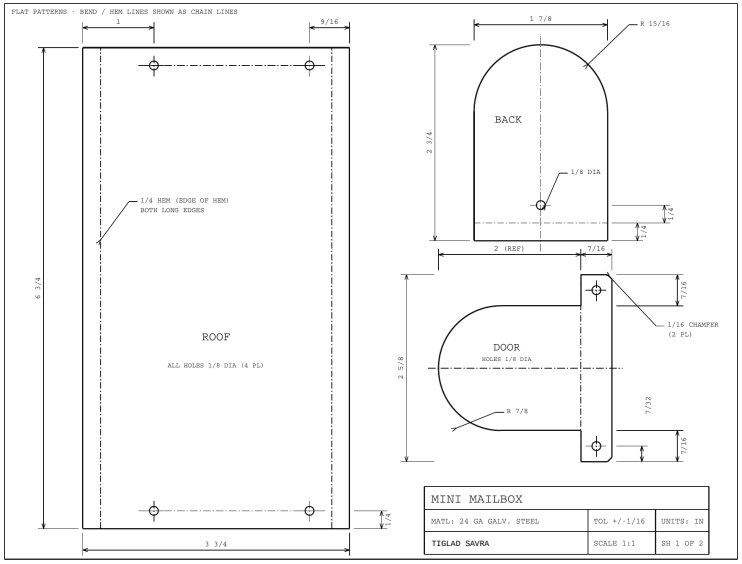
- Earned an **A** by integrating **mechanical design, soldering, circuits, RobotC,** and team troubleshooting into a functioning robot.
- Improved repeatability under time pressure by resolving sensor, motor, arm-dynamics, and alignment issues.
- Demonstrated engineering judgment by converting an open-ended prompt into a constrained, repeatable automated sequence.

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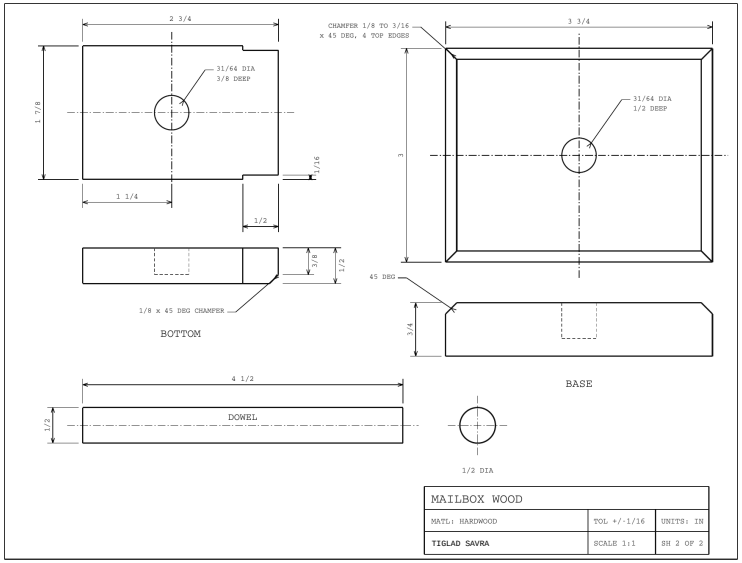
MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

MINI MAILBOX



Sheet-Metal Drawing — Roof / Door / Back



Wood Drawing — Base, Bottom & Dowel Post



Final Product

What?

- Built a miniature **hinged-door mailbox** from two subsystems: formed sheet metal and a wood support/base.
- Combined **sheet-metal fabrication** and **woodworking** to create a practical, functional mailbox.
- Produced **toleranced hand drawings** and repeatable fabrication steps for the small assembly.

How?

- Sheared, punched, filed, and arc-bent steel components to form the roof, door, and back around a matching radius.
- Cut, notched, chamfered, and drilled the wooden base and bottom using table-saw and band-saw operations.
- Aligned hinge tabs, drilled shell holes, fastened the metal body, and glued the dowel post into the base.

Results

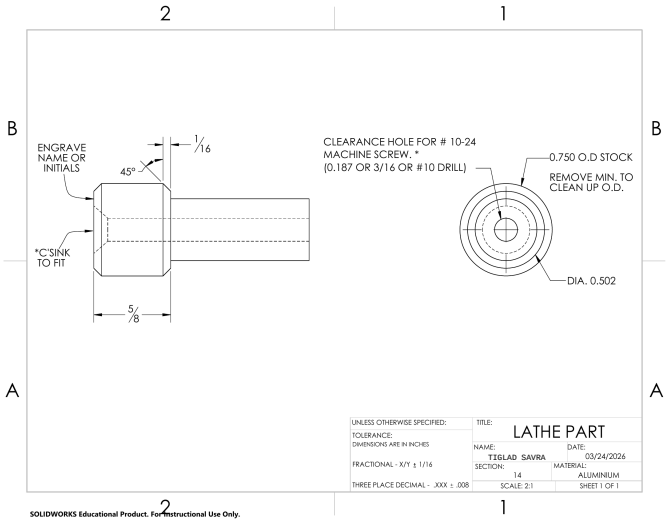
- Earned an **A** by holding dimensions to approximately **+/- 1/16 in,** including the roof bend radius and door-to-back fit.
- Assembled a **functional mailbox** with a hinged door that opens and closes.
- Identified next steps: refine the bend radius and adjust hinge clearances for smoother operation.

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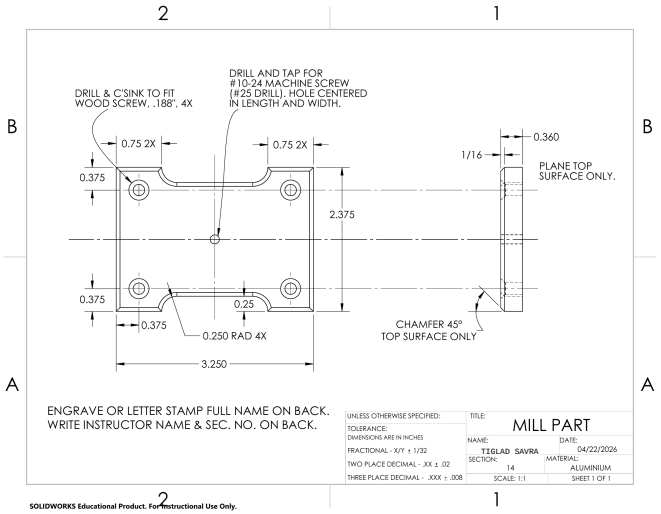
MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

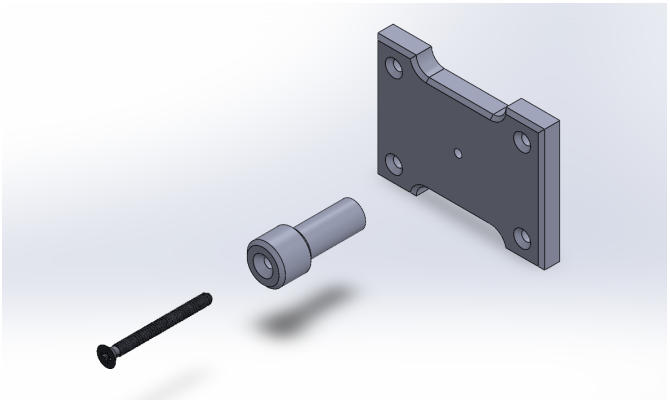
WALL MOUNTED HANGER



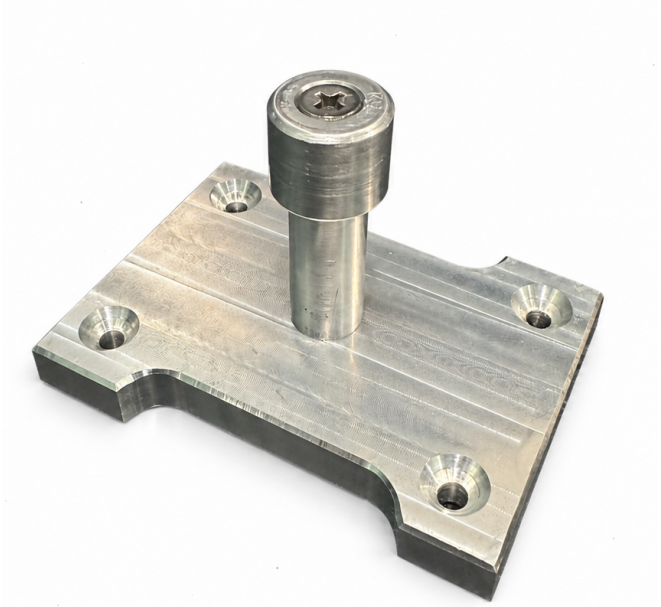
Lathe Part Drawing



Mill Part Drawing



Exploded Assembly View



Final Product

What?

- Machined an **aluminum wall-mounted hanger** from a lathe-turned peg and milled mounting plate joined by a flat-head screw.
- Designed the hanger to carry real load while maintaining a clean, presentation-ready finish on visible faces.
- Created **SolidWorks drawings** with **GD&T,** toleranced dimensions, and repeatable shop-floor fabrication notes.

How?

- Lathe: reduced stock to the shoulder diameter, faced, chamfered, drilled, countersunk, and tapped the mounting screw.
- Mill: squared 2.375 x 3.250 in stock, machined radius cutouts, drilled and countersunk holes, and tapped the mounting holes.
- Verified **thread fit** and countersink geometry, then deburred, polished, and checked the final assembly.

Results

- Earned an **A** by holding critical dimensions within a few thousandths of their target **tolerances.**
- Assembled a verified, **load-ready** wall-mounted hanger with clean screw-thread and part fit.
- Identified improvements: reduce wall thickness and add a protective coating for corrosion resistance.

TIGLAD SAVRA

MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

PERSONAL PROJECT — TOYOTA SCION FR-S X-PIPE EXHAUST MODIFICATION

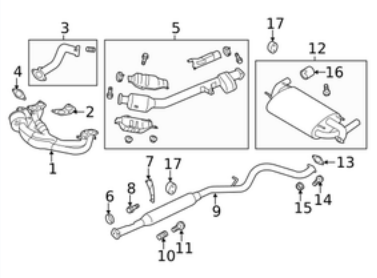


Figure 1. Exhaust-system blueprint used to identify the muffler, resonator, pipe sections, and replacement layout.



Figure 2. Before-modification view toward the rear, showing the factory muffler and tailpipe section.



Figure 3. Before-modification view farther forward near the catalytic converter and front exhaust section.



Figure 4. Removed factory muffler and connected pipe after exhaust-system disassembly.

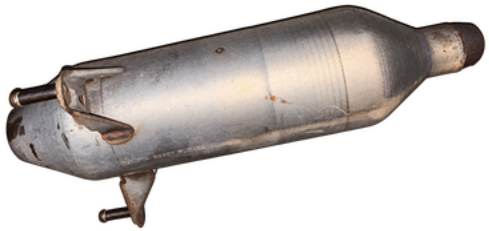


Figure 7. Removed resonator component documented during the original-system teardown.





Figures 5-6. Two views of the new copper-colored X-pipe showing fit-up and installation.

PERSONAL HANDS-ON MODIFICATION
Factory exhaust removal • X-pipe installation • welding / fit-up • flow and sound analysis

What?

- Completed a **personal project modification of a Toyota Scion FR-S exhaust system,** removing the factory muffler, resonator, and connecting pipe before installing a straighter **X-pipe** configuration.
- Used the blueprint and underbody layout to verify **part interfaces,** routing, mounting points, and clearance before reassembly.
- Supported a louder exhaust goal while learning how geometry affects flow, pulse merging, sound, and performance.

How?

- Used **drilling and sawing,** hand tools, and disassembly techniques to remove the original sections without damaging surrounding components.
- Measured and aligned replacement sections, checking **fit-up,** connections, hanger position, and underbody clearance.
- Applied **welding** concepts to secure the modification; considered heat, penetration, joint alignment, and leak prevention.
- Compared configurations to understand how an X-pipe balances pulses for smoother flow and a more aggressive sound.

Results

- Delivered a functional modification with a louder exhaust note and greater hands-on confidence in vehicle systems and fabrication.
- Learned that an X-pipe can improve pulse balance and scavenging; power gains depend on the full system, tuning, and testing.
- Strengthened **disassembly, fabrication, welding, fit-up,** and **clearance checking** skills.

TIGLAD SAVRA

MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

ME30: COMPUTER APPLICATIONS — REACTION GAME

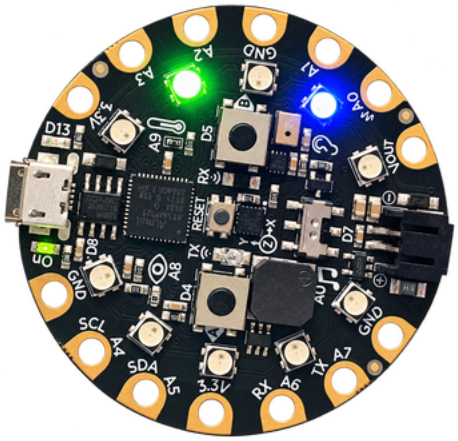


Figure 1. Adafruit Circuit Playground prototype with ten addressable LEDs, two buttons, and integrated audio feedback.

CODE EXCERPT 1 - SYSTEM SETUP

```
LED_COUNT = 10
SPIN_COLOR = (0, 0, 255)  # Blue
TARGET_COLOR = (0, 255, 0) # Green
HIT_COLOR = (0, 255, 0)
MISS_COLOR = (255, 0, 0)
SPIN_DELAY = 0.2

target = random.randint(0, LED_COUNT - 1)
cp.pixels.fill((0, 0, 0))
cp.pixels[target] = TARGET_COLOR
```

Figure 2. Python/CircuitPython setup defining LED states, feedback colors, and the initial timing condition.

CODE EXCERPT 2 - INPUT AND FEEDBACK

```
if cp.button_a or cp.button_b:
    cp.play_tone(440, 0.1)
    spinning = False

if spinner_position == target:
    SPIN_DELAY = max(0.02,
        SPIN_DELAY - 0.015)
    flash_all(HIT_COLOR)
else:
    SPIN_DELAY = 0.2
    cp.play_tone(380, 0.6)
    flash_all(MISS_COLOR)
```

Figure 3. Gameplay logic for button detection, adaptive difficulty, and success/failure feedback.

What?

- Completed a **four-week,** four-person ME30 project: a **Python/CircuitPython** reaction game on an **Adafruit Circuit Playground** with a ten-pixel LED ring, two buttons, and a speaker.
- Contributed across the **full development cycle:** architecture, **hardware integration, debugging, testing, refactoring,** and documentation.
- Turned **timing accuracy** into the control problem: a blue indicator rotates toward a randomly selected green target; the player responds with either button.

How?

- Implemented the **hardware interface** for LED colors, brightness, target selection, buttons, timing delays, and tones.
- Built a repeatable **control loop:** initialize, spin, monitor input, compare position, issue feedback, then update difficulty or reset.
- Refactored state flags and duplicate operations; reduced **SPIN_DELAY** after hits to a 0.02-second minimum and restored it to 0.20 seconds after misses.

Results

- Earned an **A** by delivering a playable prototype within the **four-week deadline,** connecting software logic to **real-time physical behavior.**
- Verified success/failure paths through **iterative testing:** correct timing increases difficulty; an incorrect response flashes red, plays the miss tone, and resets speed.
- Strengthened **Python/CircuitPython, embedded control, debugging, code organization,** and collaborative engineering skills.

TIGLAD SAVRA

MECHANICAL ENGINEER | SAN JOSE STATE UNIVERSITY

tigladsavra@gmail.com
linkedin.com/in/tiglad-savra
408-460-3822

PHYS 51: GENERAL PHYSICS II: ELECTRICITY AND MAGNETISM — WIRELESS CHARGING

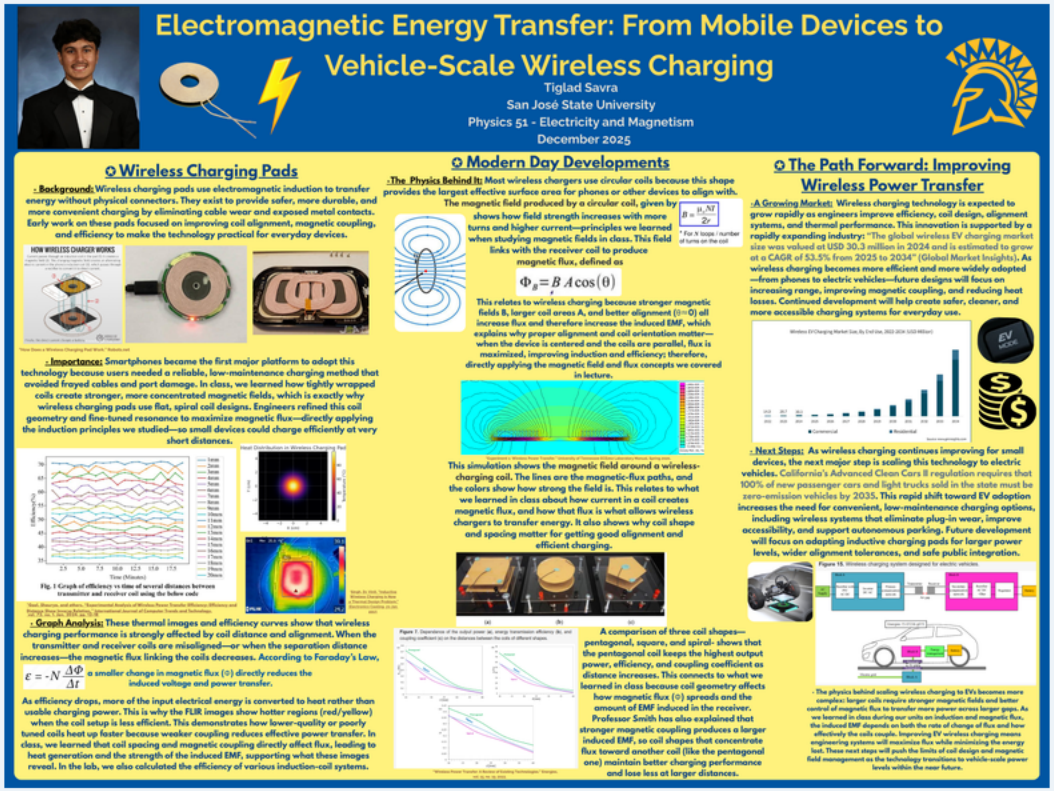


Figure 1. Final proposition poster covering wireless charging, thermal control, and vehicle-scale applications.



Figure 2. Hand-built copper induction coil used in the wireless power-transfer

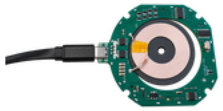


Figure 3. Wireless-charging circuit board and coil assembly with USB input.



Figure 4. Opened charger housing showing the coil, PCB, and internal

PROJECT SCOPE

Electromagnetic induction • coil design • thermal control • LED/resistor validation • EV-scale expansion

What?

- Developed a **portable wireless charger** for **Physics 51: General Physics II: Electricity and Magnetism,** applying classroom physics to a practical device.
- Extended the idea toward **vehicle-scale wireless power transfer,** where a car aligns over a charging pad instead of using a plug.
- Earned an **A** and transferred energy to a small LED/resistor load; iPhone charging was limited by the device's compatibility and protection requirements.

How?

- Built the transfer path with **coils, electromagnetic induction,** and **magnetic flux** across an air gap.
- Integrated **IR-board components,** a resistor/LED load, and thermal-management design to keep the prototype from overheating.
- Applied **Faraday's Law,** coil geometry, alignment, and **thermodynamics/heat transfer** to relate spacing and coupling to efficiency.
- Communicated the design in a research-based poster, linking the prototype to EV pads, alignment tolerance, higher power, and thermal control.

Results

- Earned an **A** by combining **electricity and magnetism, thermal management,** prototyping, and technical communication.
- Validated power transfer to an LED/resistor and identified compatibility and alignment as next-step constraints.
- Demonstrated a path to vehicle-scale charging by scaling coil area, coupling, power, and safety controls.

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linkedin.com/in/tiglad-savra
408-460-3822

ME20: DESIGN AND GRAPHICS — DOOR HANDLE ASSEMBLY

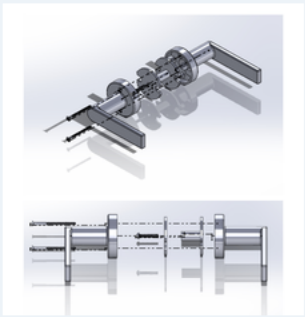


Figure 1. Exploded assembly and mating sequence showing the handle, rosettes, spindle, screws, and separate components.

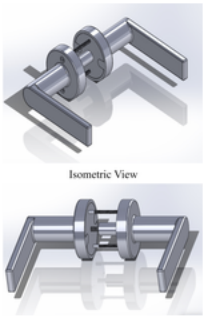


Figure 2. SolidWorks isometric views communicating the assembled form and opposing lever geometry.

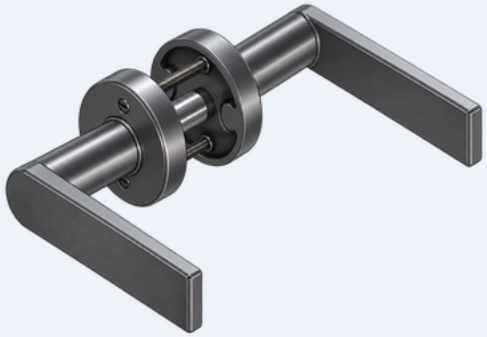


Figure 3. Realistic concept visualization of the SolidWorks assembly.

SOLO DESIGN / MANUFACTURING FOCUS

Multi-part assembly • mating • threaded fasteners • dimensional fit • IdeaMaker scale • 3D-printed door test

What?

- Designed a **multi-piece door handle** as a solo ME20 project, taking the mechanism from individual SolidWorks parts to a complete, functionally aligned assembly.
- Developed opposing levers, circular mounting plates, a central spindle/hub, and threaded fasteners so the complete handle could be assembled and installed on a door without an existing handle.
- Focused on fit, scale, and manufacturability: each part had to mate, align, and remain within dimensional constraints before production planning.

How?

- Modeled each component in **SolidWorks,** then used **assembly mates** to control the relationships across the handles, rosettes, spindle, and screws.
- Created dimensioned drawings and applied **tolerancing** and **dimensional analysis** to manage clearance, threading, alignment, and multi-part fit.
- Scaled the assembly for **IdeaMaker** and **3D printing,** then checked the printed parts for alignment and assembly as a manufacturable product.

Results

- Earned an **A** by demonstrating that the modeled assembly functioned on a handleless door, then **3D-printed** and used the finished design in an applied door test.
- Produced a complete, mechanically plausible assembly with visible fastener/threading logic and a realistic visualization of the CAD-based finished form.
- Strengthened **SolidWorks, assembly design, mating, tolerancing, technical drawings, scale control,** and mechanical problem-solving skills applicable to product design and manufacturing.